# Type MS · Hardware devices: input, output and how they connect

## Answer 1 of 4 · Question 5 part b.i · 3 marks

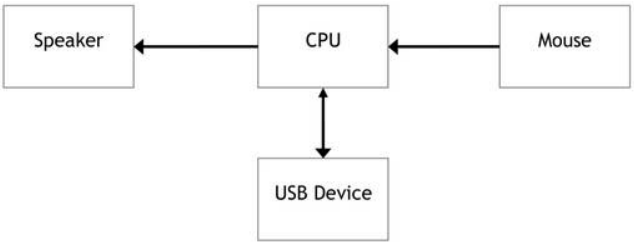
2023 JUN · Paper 01 · extract 1 of 8 · find part b.i in this mark scheme extract

<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)<br>2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)<br>4. The bee flies to its new position (1)<br>5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.<br><br>The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

## Type MS · Hardware devices: input, output and how they connect

### Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5b(i) | Award one mark for each:<br>• Line between Mouse and CPU with arrow at CPU end only (1)<br>• Line between CPU and Speaker with arrow at Speaker end only (1)<br>• Line between CPU and USB Device with arrows on both ends (1)<br>  | Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow.<br><br>Ignore any additional lines. | 3 |

# Type MS · Hardware devices: input, output and how they connect

## Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(ii) | **The only correct answer is B**<br><br>***A** is not correct because a **firewall** is an application that is loaded after the device is booted*<br><br>***C** is not correct because **the operating system** is loaded after the device is booted*<br><br>***D** is not correct because **the user interface** is loaded after the device is booted* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(iii) | Award up to **two** marks for a linked explanation, such as:<br><br>• Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1)<br><br>• Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1)<br><br>• It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1) | | 2 |

# Type MS · Hardware devices: input, output and how they connect

## Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c) | Award up to **two** marks for a linked description, such as:<br>• Uses a sensor (1) to check the water level in the machine (1)<br>• Uses a thermometer (1) to check the water temperature (1)<br>• Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1)<br>• Uses a motor (1) to rotate the drum (1) | The first mark is for identification and the second is for the linked description.<br><br>It is possible to gain one mark for a description without identifying the sensor/actuator. | 2 |

## Type MS · Hardware devices: input, output and how they connect

### Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.<br><br>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
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<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

## Type MS · Hardware devices: input, output and how they connect

### Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:<br><br>• M OR P (1)<br>• NOT C (1)<br>• <expr1> AND <expr2> (1)<br><br>Example statements:<br><br>(M OR P) AND (NOT C)<br>(M OR P) AND NOT C<br>(M AND NOT C) OR (P AND NOT C)<br>M AND NOT C OR P AND NOT C<br><br>For information only:<br><br>
<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
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<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not<br><br>Order of precedence is NOT, AND, OR<br><br>Check brackets carefully, if provided</td><td>3</td></tr>
</table>

## Type MS · Hardware devices: input, output and how they connect

### Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(e) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. The operating system uses a scheduler to control processes (1)<br>2. The operating system holds processes in a queue (1)<br>3. Some processes may be given higher priorities than others (1)<br>4. Each process gains accesses to the CPU for a short time / time slice to execute (1)<br>5. Processes are swapped to/from (the queue / CPU) | Do not award responses about memory management, as the question is about sharing the CPU only | 2 |

## Type MS · Hardware devices: input, output and how they connect

### Answer 1 of 4 · Question 5 part b.i · 3 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Hardware devices: input, output and how they connect

Answer 2 of 4 · Question 3 part a · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

| Number | | Guidance | |
|---|---|---|---|
| 3(a) | Award **two** marks for an explanation that includes hardware **and** software.<br><br>• The laptop is/contains the hardware (1) and the software allows the user to interact with it (1)<br>• Hardware gives input and output devices (1) /memory, storage, cpu, processing ability<br><br>• Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1)<br><br>• Software is the set of instructions (1) that controls the hardware (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b)(i) | Award **two** marks from:<br><br>• Lighter/thinner (1)<br>• Less heat generated (1)<br>• Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1)<br>• Silent operation (1)<br>• Less power required / longer battery life (1)<br>• More robust (due to no moving parts) (1) | | 2 |
| 3(b)(ii) | Award **two** marks for a linked explanation such as:<br><br>• Flash memory is non-volatile (1) whereas RAM is volatile (1)<br>• Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1)<br>• Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1) | | 2 |

Type MS · Hardware devices: input, output and how they connect

Answer 2 of 4 · Question 3 part a · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | Award **two** marks for a linked explanation such as:<br>• Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(d) | Award **one** mark for two correct rows.<br>Award **two** marks for four correct rows. | | 2 |

| Device | Input | Output | Both |
|---|---|---|---|
| Headset | | | ✓ |
| Mouse | ✓ | | |
| Printer | | ✓ | |
| Touch screen | | | ✓ |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e) | Award **two** marks from:<br>• Antivirus (1)<br>• Anti-spyware (1)<br>• (Software) firewall (1) | Do not award tradenames | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Hardware devices: input, output and how they connect

Answer 3 of 4 · Question 3 part d · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

| Number | | Guidance | |
|---|---|---|---|
| 3(a) | Award **two** marks for an explanation that includes hardware **and** software.<br><br>• The laptop is/contains the hardware (1) and the software allows the user to interact with it (1)<br>• Hardware gives input and output devices (1) /memory, storage, cpu, processing ability<br><br>• Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1)<br><br>• Software is the set of instructions (1) that controls the hardware (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b)(i) | Award **two** marks from:<br><br>• Lighter/thinner (1)<br>• Less heat generated (1)<br>• Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1)<br>• Silent operation (1)<br>• Less power required / longer battery life (1)<br>• More robust (due to no moving parts) (1) | | 2 |
| 3(b)(ii) | Award **two** marks for a linked explanation such as:<br><br>• Flash memory is non-volatile (1) whereas RAM is volatile (1)<br>• Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1)<br>• Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1) | | 2 |

Type MS · Hardware devices: input, output and how they connect

Answer 3 of 4 · Question 3 part d · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | Award **two** marks for a linked explanation such as:<br>• Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(d) | Award **one** mark for two correct rows.<br>Award **two** marks for four correct rows. | | 2 |

| Device | Input | Output | Both |
|---|---|---|---|
| Headset | | | ✓ |
| Mouse | ✓ | | |
| Printer | | ✓ | |
| Touch screen | | | ✓ |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e) | Award **two** marks from:<br>• Antivirus (1)<br>• Anti-spyware (1)<br>• (Software) firewall (1) | Do not award tradenames | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Hardware devices: input, output and how they connect

### Answer 4 of 4 · Question 2 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• (Touch)screen (1)<br>• Speaker (1)<br>• Vibration (motor) (1)<br>• Flashlight/LED light/torch (1)<br>• NFC chip (1) | **Accept:**<br>• display for screen<br>**Do not accept:**<br>• monitor<br>• printer<br>• headset<br>• other peripheral devices | **2** |
| Question | Answer | Additional Guidance | Mark |
| 2(b)(i) | **The only correct answer is D**<br><br>*A is not correct because cache is located closer to the CPU*<br>*B is not correct because random-access memory is a physical card on the motherboard*<br>*C is not correct because read-only memory cannot be written to* | | **1** |
| Question | Answer | Additional Guidance | Mark |
| 2(b)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• Often held/stored remotely/stored on the internet (1)<br>• Held/managed by a third party (1)<br>• Requires internet connectivity to access (1) | | **2** |

# Type MS · Hardware devices: input, output and how they connect

## Answer 4 of 4 · Question 2 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | Award **one** mark for any of the following up to a maximum of **three** marks:<br>• Clock speed/number of cycles per second (1)<br>• Number of processor cores (1)<br>• Size of cache (memory) (1)<br>• Type of cache (1)<br>• Width of data bus/word length (1)<br>• Width of address bus (1)<br>• Whether the CPU is overclocked or not (1)<br>• Pipelining architecture (1)<br>• Thermal throttling / cooling system (reducing clock speed) (1) | **Accept:**<br>• units of clock speed e.g. GHz for MP1<br>• width of bus for either MP5 or MP6 but not both<br>**Do not accept:** cache without clarification of size or type | 3 |
| Question | Answer | Additional Guidance | Mark |
| 2(c)(ii) | Award up to **two** marks for a linked description such as:<br>• The CPU places an address on to the address bus (1) to specify the location in main memory (1)<br>• (The address bus) contains the location in RAM (1) to read from/write to (1)<br>• The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1)<br>• The address bus carries the memory location (1) of the current/next instruction (1) | Explanation can be a combination of different marking points. E.g. MP1a and MP2b | 2 |
| Question | Answer | Additional Guidance | Mark |
| 2(d) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• Sub-atomic particles (1)<br>• Qubit(s) (1)<br>• Entanglement (1) behaviour of a pair of qubits can influence each other<br>• Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously | **Do not accept:**<br>'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question | 2 |
| | | **Total for question 2** | **12** |

## Type MS · Hardware devices: input, output and how they connect

### Answer 4 of 4 · Question 2 part a · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|